Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Databricks Application Development, Agentic AI Workflows, LLM Prompt Orchestration, MLOps & Model Lifecycle Management, Scalable ETL & Data Pipelines, Autonomous Anomaly Detection, Time-Series Forecasting (SARIMA), Dimensional Modeling (Star/Snowflake, SCD Type 1/2), Data Governance & Lineage (Collibra), Executive BI & Semantic Modeling

Programming Languages & Tools: Databricks, Azure (ADF, Databricks, Synapse, ADLS, Azure SQL), Snowflake, AWS (S3, Redshift), Apache Spark, PySpark, Python, SQL, HTML, Power BI (DAX), Tableau, XGBoost, Scikit-Learn, KIRO IDE, Copilot, Collibra

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Lead Cloud Data & AI Engineer — **May 2022 – Present**

- **Architected production-grade Databricks applications and autonomous agentic workflows** (utilizing KIRO IDE, prompt orchestration, and Copilot) to automate schema discovery, data profiling, and code migration, reducing manual delivery turnaround by 40-50% while seamlessly connecting enterprise data, AI capabilities, and business workflows.
- **Engineered and operationalized autonomous machine learning anomaly detection** frameworks identifying schema variations, distribution drift, and ETL discrepancies with 95% precision, implementing robust MLOps practices to safeguard downstream BI and pipeline integrity.
- **Designed and scaled high-throughput cloud data pipelines and ETL workflows** across Azure (ADF, Databricks, Synapse, ADLS, Azure SQL) and Snowflake, utilizing Apache Spark / PySpark to crunch hundreds of terabytes while maintaining 99.9% data availability SLAs and adhering to strict governance standards.
- **Built and deployed predictive machine learning models at scale** (XGBoost, Random Forests, Scikit-Learn) and advanced time-series models (SARIMA) to forecast customer behavior and field support demand, delivering actionable insights that optimized workforce allocation and improved reliability by 20%.
- **Engineered automated testing and evaluation frameworks for conversational AI** (SpectrumIQ enterprise virtual assistant), cutting manual QA turnaround by 50% and accelerating production release cycles while ensuring high reliability and performance.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Developed and deployed production-ready predictive classification models** using Scikit-Learn, XGBoost, and Python, transforming complex transactional telemetry into reliable behavioral forecasts.
- **Implemented rigorous data preparation and feature engineering processes** to support large-scale machine learning workloads, integrating disparate sources to drive robust enterprise analytics.
- **Collaborated with cross-functional technical stakeholders** to translate business requirements into high-performance machine learning designs and scalable data applications.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Engineered executive Key Operating Metric (KOM) dashboards** and semantic models in Power BI (DAX) and Tableau to translate complex transactional telemetry into actionable corporate strategy.
- **Built source-to-target reconciliation firewalls and Collibra data lineage** frameworks, reducing operational turnaround by over 80% and ensuring strict data integrity and governance compliance.
- **Optimized enterprise data access and transformation using advanced SQL** and dimensional modeling (Star/Snowflake schemas, SCD Type 1/2) to support mission-critical reporting workloads.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture **Pune, India**

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.